Autonomous Hypertension Management

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Design Principles

1. Do no harm
2. Deterministic by default
3. Diverse safeguards
4. Explicit scope boundaries
5. Verification before intensification
6. Every decision is traceable
7. Earned autonomy
8. Clinician authority

Assumptions

- Adult patients only (age 18+)
- Primary hypertension
- Patients already diagnosed by a clinician
- English-speaking patients
- Non-pregnant patients
- Outpatient setting
- Appropriate regulatory authorization

Clinical Problems

- Identifying and classifying hypertension
- Selecting appropriate antihypertensive therapy
- Titrating medications to achieve BP goals
- Calculating cardiovascular risk to guide treatment intensity
- Detecting and managing hypertensive emergencies vs severe hypertension
- Ensuring medication adherence
- Managing medication side effects
- Monitoring for drug-related adverse effects
- Identifying resistant hypertension
- Screening for secondary hypertension

Safety Failure Modes – 5 most likely causes of severe harm

Failure	Defenses	Deterministic	Residual
Missed emergency	5 layers	3/5	LOW
Overtreatment → fall	6 layers	3/6	MODERATE
Wrong drug in pregnancy	6 layers	3/6	LOW
Electrolyte abnormality post-uptitration/new drug	6 layers	6/6	VERY LOW
Missed secondary HTN	6 layers	4/6	MODERATE

Architecture

Main agent - LLM

- Conversation
- Tool calling
- Clinical education
- Symptom recognition
- Adherence assessment
- Side effect mapping
- Emergency response
- Reproductive safety screening
- Lifestyle coaching
- Interpret ambiguity
- Conversation scope enforcement

Data Stores

- EHR
- Vector
- Operational
- Reference tables

System Orchestrator

- State Machine
- Decision Trees
- RAG Reasoning
- Safety Pipeline

Other

- Ingestion services: Withings, Surecripts, OpenFDA HealthGorilla, RxNorm, external clinical notes
- Pre and post processing layers (LLM + rules)
- LLM-as-judge
- Note generator
- Session summarizer
- Follow-up scheduler
- Clinician dashboard

Patient Context Injection

```
## Current Patient Context
    Name: Jane Smith | Age: 58 | Sex: F
    Care State: AWAITING_DRUG_EFFECT (since
2026-01-28, 15 days)
    BP Goal: <130/80
    Current Medications:
    - Lisinopril 20mg daily (ACEi, high
dose) — started 2024-03-15
    - Amlodipine 5mg daily (DHP-CCB, low
dose) — started 2026-01-28 ← NEW

    Last 3 BP Readings:
    - 2026-02-10: 142/88 (home)
    - 2026-02-07: 138/86 (home)
    - 2026-02-03: 151/94 (home)

    Adherence: Low risk (screener 0, fills
on schedule)
    CVD 10yr (PREVENT): 14.2%

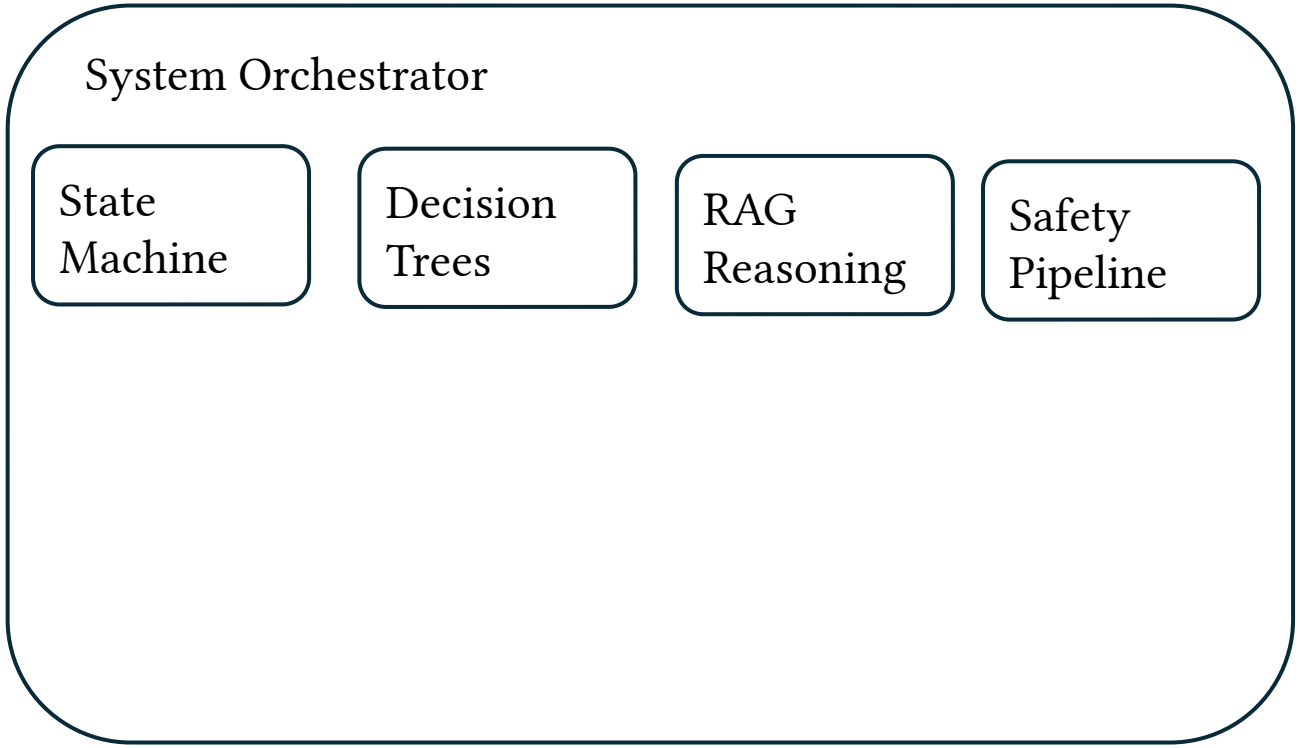
    Pending Actions:
    - Follow-up BP check due: 2026-02-25 (4
weeks post amlodipine start)
    - Recheck labs (K+, Cr) due: 2026-02-11
← OVERDUE

    Active Alerts:
    - Lab recheck overdue (1 day)
    Conditions: Essential HTN, Type 2 DM


    Allergies: None documented
```

Tools

- Agent uses tools, the results of the tool use are used by the system orchestrator to handle state machine changes, tree traversal, RAG reasoning, and safety pipeline



Tools - Deterministic

- record_bp_reading
  - validate_bp_reading
  - add_bp_observation
  - classify_bp(systolic, diastolic)
  - db.get_patient_state **--- retrieves from state machine**
  - fhir.get_bp_goal
  - fhir.get_recent_bps("pt-123", limit=3)
  - is_at_goal(recent_readings, bp_goal, min_readings=3)
  - if current_state == "AWAITING_DRUG_EFFECT":
  - days_since_change = (now() - state_metadata["therapy_change_date"]).days
    - if at_goal and days_since_change >= 14:
  - new_state = "AT_GOAL" **--- state transition**
    - elif not at_goal and days_since_change >= 28:
  - new_state = "CONFIRMING_UNCONTROL"

Patient State


```
{
    "state":
"AWAITING_DRUG_EFFECT",
    "entered_at": "2026-02-
15T14:30:00Z",
    "trigger": "therapy_change",
    "context": {
        "therapy_change_date": "2026-
02-15",
        "changed_medications":
["lisinopril 10mg -> 20mg"],
        "decision_tree_run_id": "dt-
run-42",
        "follow_up_due": "2026-03-15",

        "readings_since_change": 3
    }
}
```

Tools (additional)

Side Effect Management

- assess_symptom

Therapy

- evaluate_therapy
- execute_prescription
- stop_medication

Orders

- order_labs

Data Retrieval

- get_patient_summary
- get_bp_history get_lab_results
- get_medication_details

Knowledge

- search_guidelines
- check_drug_interaction

Escalation

- escalate
- send_to_clinician

Adherence

- administer_adherence_screener

Scheduling

- schedule_follow_up

State

- update_care_state

Screening

- flag_secondary_htn_clue

Agent cannot

1. Directly write to FHIR (must go through tools)
2. Skip safety checks (execute_prescription validates them)
3. Stop a medication autonomously (always clinician approval)
4. Override a clinician rejection
5. Prescribe outside the recommendation pipeline
6. Change care state directly (only via update_care_state with specific update types)
7. Access other patients' data
8. Delete clinical records

LLM Roles

Role	Model	Reasoning
Main Agent	GPT-4o	Best tool-calling, fast, empathetic tone, reliable function calling. Patient-facing. Focused prompt
LLM-as-Judge	Claude Sonnet 4.5	DIFFERENT PROVIDER for model diversity. If GPT-4o has a blind spot, Claude may catch it. Strong structured evaluation. Not speed-critical.
RAG Reasoner	GPT-4o	Same model as agent (shares prompt patterns). Strong citation grounding. Separate system prompt.
Note Generator	GPT-4o-mini	Structured input -> structured output. Template-guided. Runs async. Lower cost for high volume.
Intake Classifier	GPT-4o-mini	Simple classification task. Fastest, cheapest.
Scope Classifier	GPT-4o-mini	Classifies every inbound message as bp_related / emergency / off_topic / ambiguous. Runs before main agent. Fast, cheap, high-volume.
Tone Validator	GPT-4o-mini	Validates response tone against care-state expectations. Low volume (~5-10% of responses).

Message Processing Pipeline

PATIENT MESSAGE ARRIVES

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PRE-PROCESSING LAYERS (before main agent)

1. Emergency keyword scan (deterministic)
2. Scope & intent classifier
3. Patient context injection
4. Secondary HTN symptom pre-scan (and nudge if necessary)

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MAIN AGENT (responses and tool calls)

↓

SYSTEM ORCHESTRATOR (state machine, trees, safety pipeline)

↓

POST-PROCESSING LAYERS (after agent response, before delivery to patient)

5. Tone validator
6. Scope re-check
7. Emergency re-scan
8. PHI / proprietary boundary check

↓

RESPONSE

Context

Layer	Content	Size	Always included?
System prompt	Agent persona, capabilities, instructions, scope rules	~2,000 tokens	Yes
Patient context	Injected structured context	~500 tokens	Yes
Current session	All messages in the current session	Variable	Yes
Recent history summary	AI-generated summary of the last 2-3 sessions	~500 tokens	Yes
Tool results	Results from tool calls in the current turn	Variable	Yes
Care thread summary	AI-generated summary of the full care relationship (updated monthly or after major events)	~800 tokens	If > 5 sessions exist

State Machine

State determines:

- Which decision tree to use
- Which follow-up schedule to use
- How to interpret new BP readings

Lifestyle only	Confirming uncontrolled	Therapy adjustment	Awaiting drug effect	At goal
Side effect eval	Confirming uncontrolled	Adherence Intervention	Therapy adjustment	Awaiting drug effect
Possible overtreatment	Crisis	Pending clinician	Resistant hypertension	

State Machine – Transitions – new patient and stable states

From	To	Trigger
NEW_PATIENT	LIFESTYLE_ONLY	Stage 1 HTN + PREVENT <7.5%
NEW_PATIENT	THERAPY_ADJUSTMENT	Stage 2 HTN or PREVENT ≥7.5%
NEW_PATIENT	AT_GOAL	Already on meds + BP at goal
NEW_PATIENT	CONFIRMING_UNCONTROL	Already on meds + BP above goal
NEW_PATIENT	CRISIS	BP ≥180/120 or emergency symptoms
Stable states		
LIFESTYLE_ONLY	THERAPY_ADJUSTMENT	Still elevated after 3-6 months
LIFESTYLE_ONLY	AT_GOAL	Lifestyle modifications worked
AT_GOAL	CONFIRMING_UNCONTROL	BP rises above goal
AT_GOAL	SIDE_EFFECT_EVAL	Side effect reported
AT_GOAL	POSSIBLE_OVERTREATMENT	SBP <100 or hypotensive symptoms

State Machine – Transitions – assessment and treatment

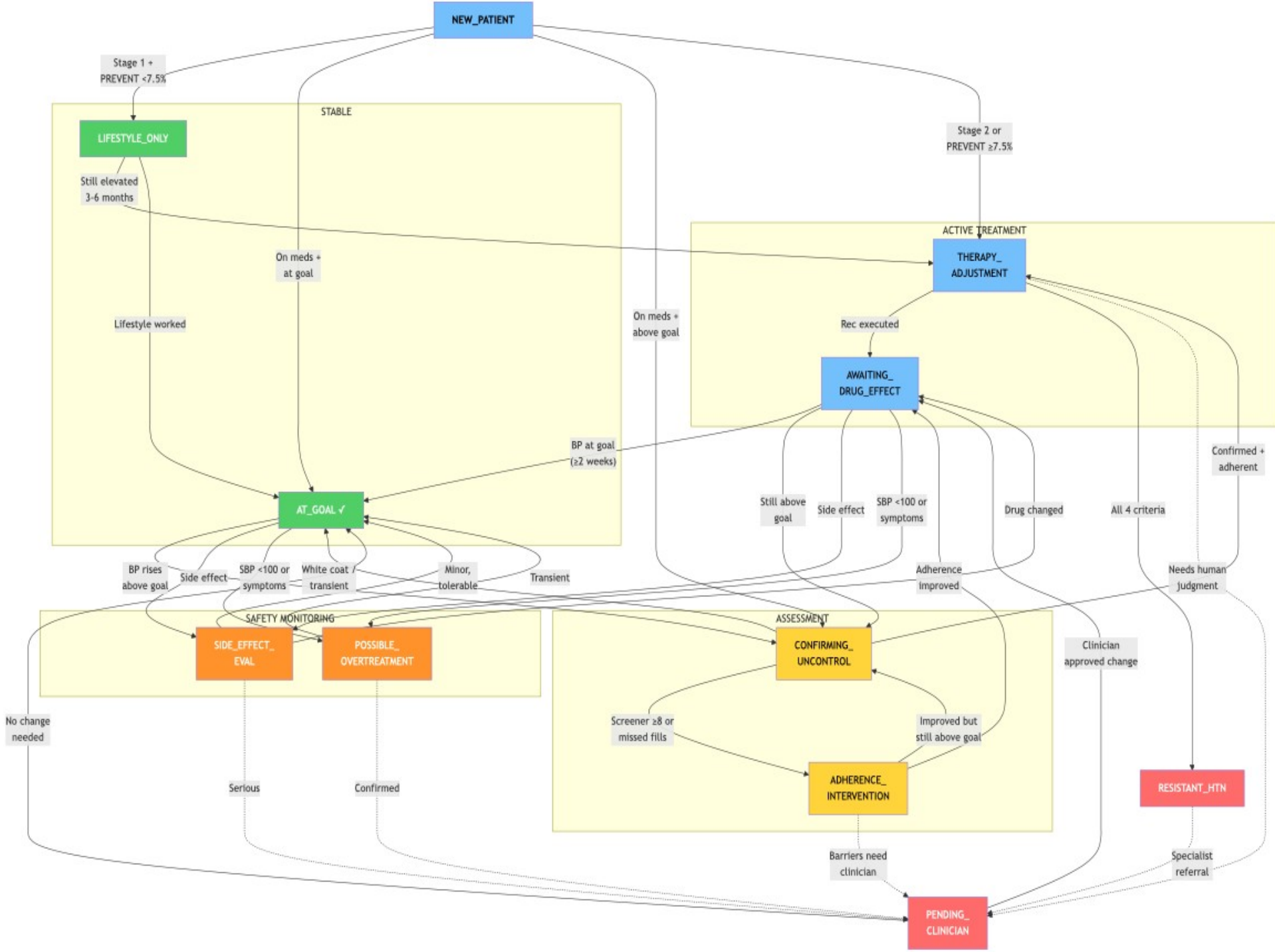
From	To	Trigger
Assessment		
CONFIRMING_UNCONTROL	AT_GOAL	White coat effect / transient spike
CONFIRMING_UNCONTROL	ADHERENCE_INTERVENTION	Screener ≥8 or missed pharmacy fills
CONFIRMING_UNCONTROL	THERAPY_ADJUSTMENT	Confirmed uncontrolled + adherent
ADHERENCE_INTERVENTION	AWAITING_DRUG_EFFECT	Adherence improved
ADHERENCE_INTERVENTION	CONFIRMING_UNCONTROL	Improved but BP still above goal
ADHERENCE_INTERVENTION	PENDING_CLINICIAN	Barriers need clinician involvement
Active treatment		
THERAPY_ADJUSTMENT	AWAITING_DRUG_EFFECT	Recommendation executed
THERAPY_ADJUSTMENT	PENDING_CLINICIAN	Needs human judgment (Layer 3)
THERAPY_ADJUSTMENT	RESISTANT_HTN	All 4 resistant HTN criteria confirmed
AWAITING_DRUG_EFFECT	AT_GOAL	BP at goal (≥2 weeks post-change)
AWAITING_DRUG_EFFECT	CONFIRMING_UNCONTROL	Still above goal after drug effect period
AWAITING_DRUG_EFFECT	SIDE_EFFECT_EVAL	Side effect reported
AWAITING_DRUG_EFFECT	POSSIBLE_OVERTREATMENT	SBP <100 or hypotensive symptoms

State Machine – Transitions – safety and escalation

From	To	Trigger
Safety monitoring		
SIDE_EFFECT_EVAL	AWAITING_DRUG_EFFECT	Drug changed / switched
SIDE_EFFECT_EVAL	AT_GOAL	Minor side effect, tolerable
SIDE_EFFECT_EVAL	PENDING_CLINICIAN	Serious side effect
POSSIBLE_OVERTREATMENT	PENDING_CLINICIAN	Confirmed overtreatment, downtitrate
POSSIBLE_OVERTREATMENT	AT_GOAL	Transient, readings normalized
Exception / escalation		
RESISTANT_HTN	PENDING_CLINICIAN	Specialist referral
PENDING_CLINICIAN	AWAITING_DRUG_EFFECT	Clinician approved therapy change
PENDING_CLINICIAN	AT_GOAL	Clinician determined no change needed

State Machine – Transitions – crisis

From	To	Trigger
Crisis (universal — from any state)		
LIFESTYLE_ONLY	CRISIS	BP ≥180/120 or emergency symptoms
AT_GOAL	CRISIS	BP ≥180/120 or emergency symptoms
CONFIRMING_UNCONTROL	CRISIS	BP ≥180/120 or emergency symptoms
AWAITING_DRUG_EFFECT	CRISIS	BP ≥180/120 or emergency symptoms
SIDE_EFFECT_EVAL	CRISIS	Dangerous reaction
POSSIBLE_OVERTREATMENT	CRISIS	Syncope or fall
CRISIS	PENDING_CLINICIAN	Always (immediate clinician notification)
CRISIS	AWAITING_DRUG_EFFECT	After crisis resolved + therapy adjusted


NEW_PATIENT
Stage 1 + PREVENT <7.5%
STABLE
LIFESTYLE_ONLY
Still elevated 3-6 months
Lifestyle worked
On meds + at goal
Stage 2 or PREVENT ≥7.5%
ACTIVE TREATMENT
THERAPY_ ADJUSTMENT
On meds + above goal
Rec executed
AWAITING_ DRUG_EFFECT
BP at goal (≥2 weeks)
Confirmed + adherent
AT_GOAL ✓
Still above goal
Side effect
SBP <100 or symptoms
Drug changed
All 4 criteria
BP rises above goal
Side effect
SBP <100 or symptoms
White coat / transient
Minor, tolerable
Transient
Adherence improved
Needs human judgment
SAFETY MONITORING
SIDE_EFFECT_ EVAL
POSSIBLE_ OVERTREATMENT
ASSESSMENT
CONFIRMING_ UNCONTROL
Clinician approved change
No change needed
Screener ≥8 or missed fills
Improved but still above goal
Serious
Confirmed
ADHERENCE_ INTERVENTION
RESISTANT_HTN
Barriers need clinician
Specialist referral
PENDING_ CLINICIAN

Decision Tree Selection

1. **Care state selects the entry point**

   Initial therapy, uncontrolled monotherapy, or uncontrolled combination therapy.

2. **Verify the clinical context**

   Use structured observations, medication history, adherence, and relevant laboratory data.

3. **Resolve each decision node**

   Record the question, answer, source data, and confidence for the audit trail.

4. **Escalate unresolved decisions**

   Missing data or clinical judgment routes to retrieval with citations or clinician review.

From Tree Output to a Proposed Prescription

1. **Select a drug class**

   The tree returns a class-level recommendation and the evidence supporting its path.

2. **Resolve the medication and dose**

   Reference tables apply allergies, renal function, prior tolerance, and formulation constraints.

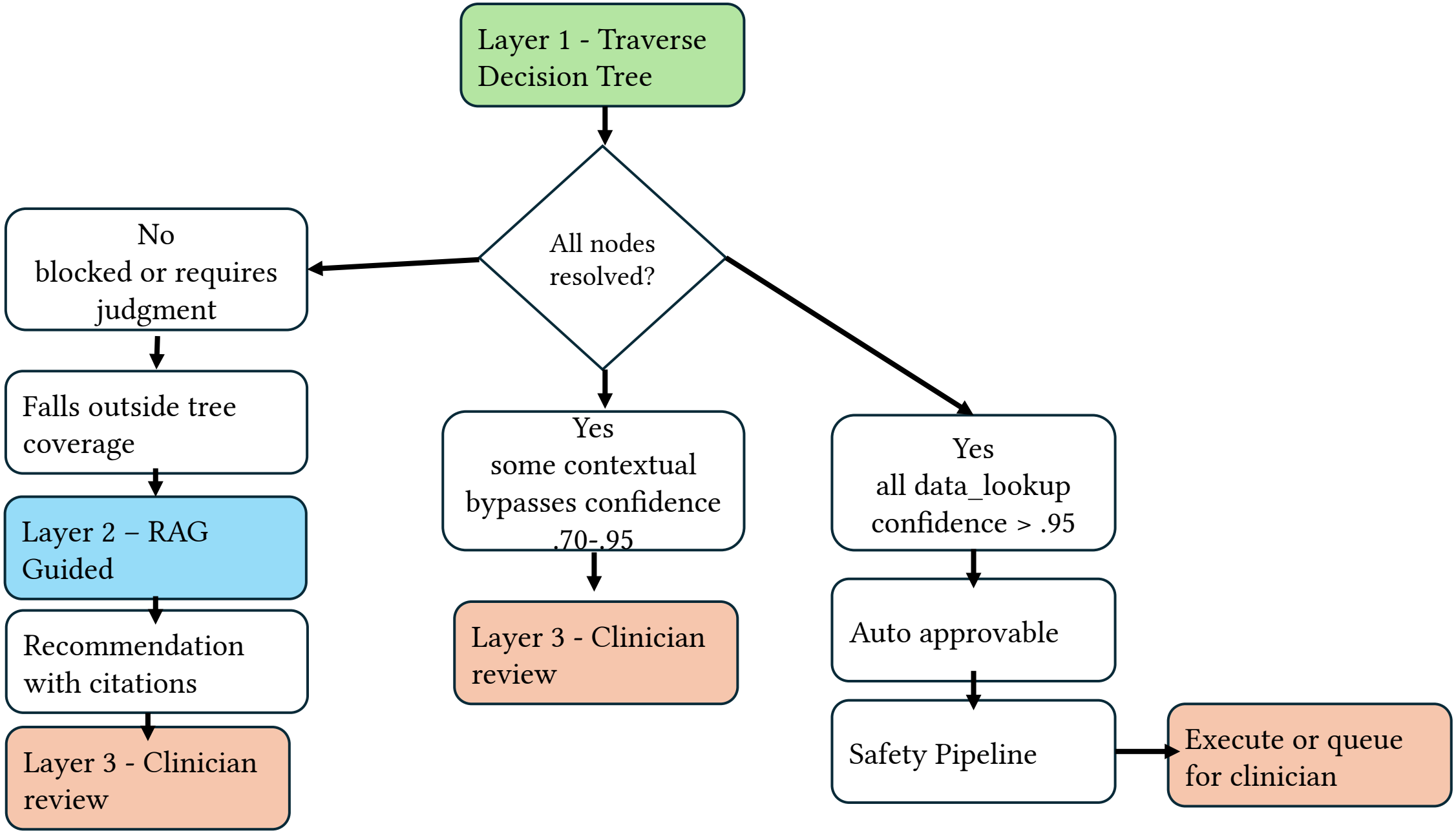
3. **Run the safety pipeline**

   Deterministic checks and a separate model review assess the proposed action.

4. **Apply the authorization policy**

   Required clinician approval precedes execution. The system records the decision and schedules follow-up.

3 Layer Clinical Decision Model



RAG Uses – patient and clinical

Patient communication

- Patient education ("What does Stage 2 hypertension mean?")
- Side effect counseling ("I've been having swollen ankles")
- Lifestyle coaching ("What dietary changes help?")
- Drug explanations ("Why was I started on this drug?")
- Decision justification ("Why this drug instead of another?")

Clinical reasoning

- Side effect-driven drug changes (e.g. no tree covers "controlled but needs to switch")
- Lab-driven dose adjustments ("K+ is 5.1 on ACEi, what to do?")
- Comorbidity-specific nuances (diabetes + thiazide metabolic effects)

RAG Constraints

- Layer 2 recommendations ALWAYS go to clinician review -- never autonomous
- LLM can only cite from uploaded guidelines, not training data
- All citations are verified against retrieved chunks before output
- Safety layer still runs on Layer 2 recommendations
- Audit trail captures exact quotes the AI relied on + retrieval metadata
- If guidelines don't address a scenario, AI says "insufficient guidance" (Layer 3)

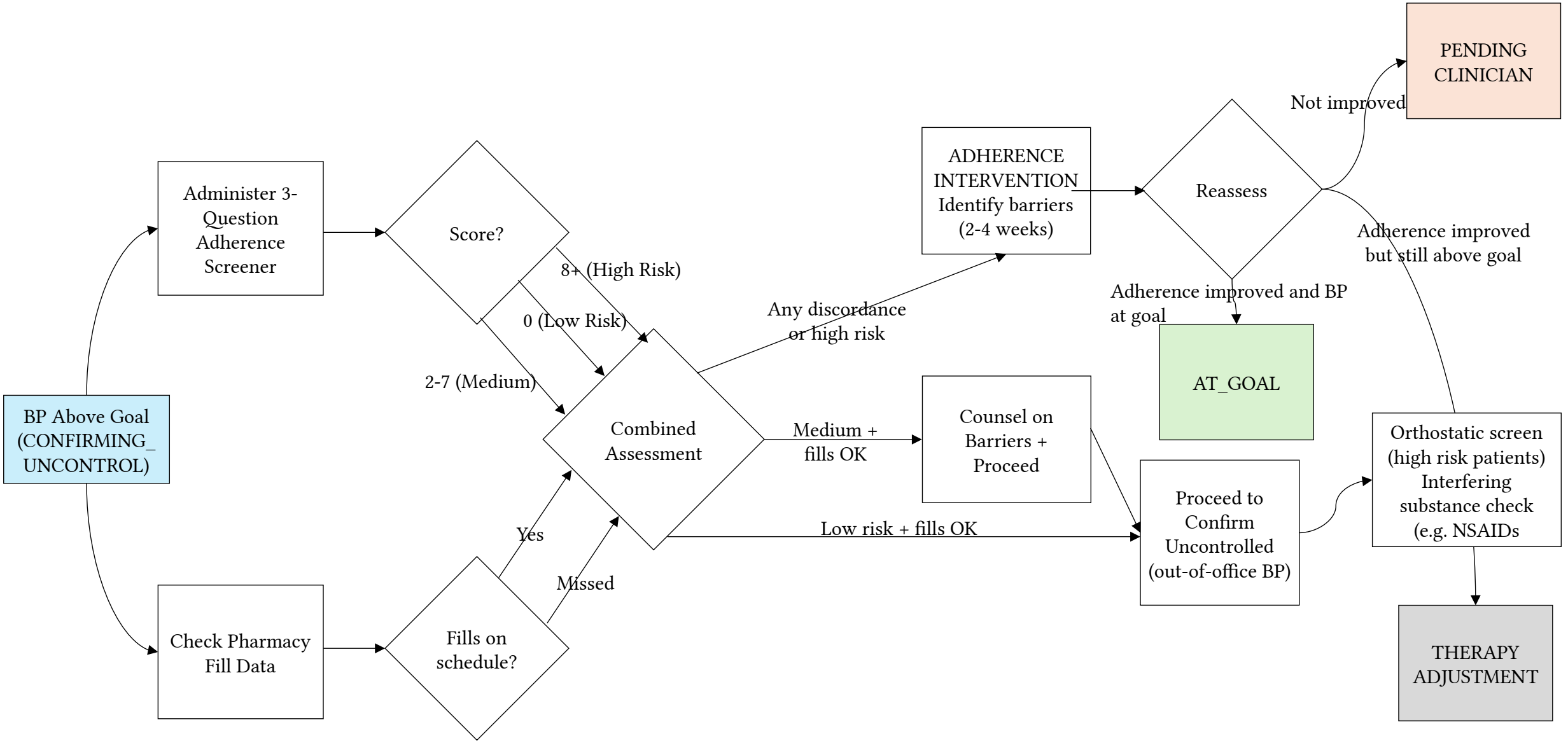
Safety Layer – deterministic checks + LLM-as-judge

1. Dangerous combination (e.g. ACEi + ARB)
2. Allergy (matches proposed drug and class against allergy list)
3. Contraindications (e.g. severe asthma for BB)
4. Lab safety (K, GFR, hCG)
5. Hypotension and orthostatic risk (recent readings, fall risk)
6. Side effect history (matches proposed against intolerance list)
7. Drug interactions (e.g. thiazide + lithium)
8. Dose ceiling check
9. Renal dosing adjustment (e.g. if GFR < 30 start ACEi at lowest dose)
10. LLM-as-judge
    - Approve, caution, reject
    - Receives patient context, proposed action, decision tree path and/or RAG citations, and results of all 9 preceding safety checks)
    - Model from different vendor

- Failure 1: Missed emergency (hypertensive emergency, chest pain)

Defense layer	Mechanism	Catches	Can fail?
Pre-processing layer: Emergency keyword detection	Hard-coded keywords/phrases ("chest pain", "can't breathe", "worst headache", "can't see", etc.) run on every inbound message, ~0ms, no LLM dependency	Direct, unambiguous emergency language	Yes — patient uses non-standard wording ("my chest feels heavy and weird")
Main agent: LLM classification	GPT-4o evaluates messages that don't trigger keywords. Trained to ask targeted follow-ups for ambiguous messages.	Nuanced, indirect, or partial descriptions	Yes — LLM could misclassify or fail to ask the right follow-up
Post-processing layer: Emergency re-scan	Same keyword scan applied to agent's response. If agent's reasoning mentions emergency indicators, catches them even if the agent didn't escalate.	Agent identifies danger in its reasoning but fails to act on it	Yes — only catches keyword matches in the response text
BP reading classification	Deterministic: any reading ≥180/120 → crisis pathway regardless of what the patient says	Numerically dangerous readings even when patient reports "feeling fine"	No — deterministic, cannot fail if reading is recorded
Safety invariant #1	Emergency keyword scan works even when GPT-4o, Claude, FHIR, and all external services are down	Total system degradation	No — keyword list is in-memory, zero external dependencies

Failure 2: Overtreatment (mitigated by adherence gating)



Data Stores

EHR (FHIR)

Patient | Condition | MedicationStatement | MedicationDispense | AllergyIntolerance | Observation | Goal | Composition | DocumentReference | Care Plan

Read access: Orchestrator, Agent (via tools like get_bp_history)

Write access: Agent **tools** (e.g. record_bp_reading), ingestion services (e.g. Surescripts), clinicians

Vector

Chunked clinical guidelines read by RAG reasoner

- AHA 2025
- UpToDate

Operational

Patient care state | conversation history (e.g. message, tool call, tool result) | decision audit log (e.g. tree traversals) | escalations | follow-up schedules | RAG reasoning logs

Read access: Orchestrator, Agent (care thread summaries injected into context)

Write access: Orchestrator (e.g. state transitions), session summarizer

Reference Tables

Lookup tables that encode clinical rules

(separates clinical knowledge from code)

Lookup Table Example

Drug	Class	HTN Start	Titration Steps	Max HTN	Frequency	Tier Map
Lisinopril	ACEi	10mg	10 → 20 → 40	40mg	Once daily	L=10, M=20, H=40
Enalapril	ACEi	5mg	5 → 10 → 20 → 40	40mg	Twice daily	L=5, M=10-20, H=40
Losartan	ARB	50mg	50 → 100	100mg	Once daily	L=50, H=100
Valsartan	ARB	80mg	80 → 160 → 320	320mg	Once daily	L=80, M=160, H=320

Additional tables:

- Drug interactions
- Renal dosing adjustments
- Disease-specific indications
- Single pill combinations
- Side effects
- BP goal rules
- HBPM protocol parameters

Core Data Flows (examples)

Flow 1: Patient submits BP reading

- Parse -> validate -> store in FHIR (with position tag if orthostatic) -> classify BP -> assess pulse -> check state machine -> (possibly trigger tree traversal -> safety checks -> execute or queue) -> generate note -> respond to patient

Flow 2: Lab result arrives

- Health Gorilla -> normalize to FHIR Observation -> immediate lab safety check against patient's current drugs
  - -> CRITICAL: auto-escalate + hold medication + notify clinician + message patient. WARNING: Layer 2 reasoning + queue for clinician.
  - -> NORMAL: log, inform patient if follow-up lab.
- Pregnancy detection: If lab is HCG AND result is positive/elevated AND patient is on ACEi/ARB ⟶ treat as CRITICAL

Flow 3: Side effect report

- Patient message -> symptom extraction -> structured side effect lookup (match against current meds) -> RAG counseling narrative -> ask follow-up questions -> if drug change needed: Layer 2 reasoning -> clinician review

Flow 4: Follow-up timer

- Scheduler fires daily
  - > AWAITING_DRUG_EFFECT + follow-up due -> proactive message
  - AT_GOAL + routine check due -> request reading
  - LIFESTYLE_ONLY + trial end -> re-evaluate
  - Monitoring gaps -> prompt labs
  - PENDING_CLINICIAN > 48 hours -> nudge clinician

Trade-offs

Chose	Over	Because	But
Single agent + policy layers	Multi-agent	Handoff failures are a safety risk	More infrastructure than monolithic single agent, more work to add capabilities
Deterministic trees (80%)	Pure LLM reasoning	Prescribing must be auditable	Overhead of building and maintaining the trees
Cross-provider judge	Same-provider	Model diversity catches correlated failures	Increase in infrastructure, latency
Adherence gating	Escalate immediately	#1 cause of uncontrolled BP is non-adherence	More complicated state machine, slower uptitration
Two-tier emergency	LLM-only	Keywords work when LLM is down	Keyword table to maintain
Graduation path	Start autonomous	Trust built incrementally	More work to assess performance, need to remediate if necessary

Autonomy is earned (prescribing)

Phase 1
All clinician approval

>250 recs
- 95% approval
- 0 safety rejections
- Judge agreement > 90%

Phase 2
Uptitrate = Autonomous

>500 total
>100 uptitrations
- 95% add drug approval
0 safety rejections
> Judge agreement > 90%

Phase 3
Uptitrate + add = Autonomous

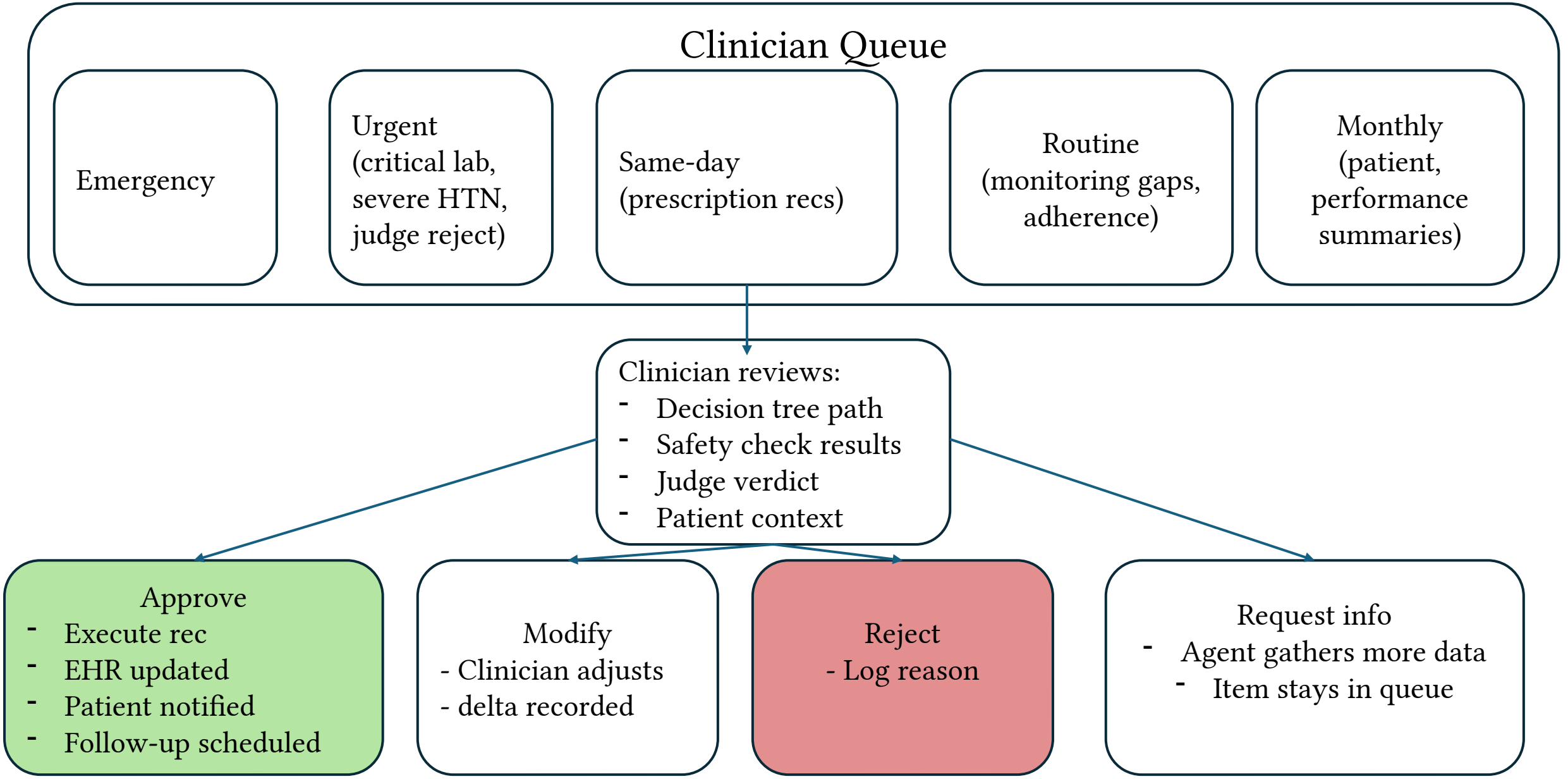
Autonomy is earned (other)

- Record BP reading
- Classify BP (deterministic rules)
- Lifestyle coaching
- Trigger crisis escalation (hard-coded safety layer)
- Schedule follow-up (guideline-based intervals)
- Order routine monitoring labs (per drug class rules)
- Flag monitoring gaps (deterministic)
- Patient education (RAG over guidelines)

Final Levels of autonomy

- Remove a drug
- Down-titrate a drug
- Switch drug due to side effect
- Resistant HTN management
- Crisis management

Clinician Dashboard



Follow-up based on care state (sample)

Care state	Follow-up interval	What happens at follow-up	Escalation if missed
AWAITING_DRUG_EFFECT	Monthly (typically 4 weeks) after therapy change	Request BP readings. Ask about side effects. Check if readings are trending toward goal. Per 2025 AHA/ACC Section 5.4 Rec 4: "follow-up at monthly intervals until control is achieved" (COR 1, B-R).	Proactive outreach at due date. If no response within 3 days, flag for care team.
AT_GOAL	3 months (routine)	Request BP check. Brief symptom screen. Confirm medication adherence. Review any new concerns.	Proactive outreach. If no response, try again in 1 week. After 2 attempts, flag for care team.
LIFESTYLE_ONLY	3 months initially, extend to 6 months if stable	Request BP check. Review lifestyle progress. Re-evaluate whether medication is needed.	Same as AT_GOAL.
CONFIRMING_UNCONTROL	4-10 days (HBPM protocol)	Collect ≥12 home readings over ≥3 days (first day discarded). Administer adherence screener. Cross-reference pharmacy fills. Orthostatic screen if high-risk. Day 3 and 7 reminders if insufficient readings. Day 10 timeout → clinician flag + ABPM referral.	Important — uncontrolled BP should not sit unaddressed. Flag for care team after Day 10 timeout.

Lab Monitoring Follow-Up

Two types:

- 1) Event based gaps (e.g. by medication start, dose increase)
- 2) Interval-based gaps

Trigger	Labs to order	When	Follow-up
New ACEi/ARB/MRA started or dose increased	K+, Creatinine, eGFR	Baseline (if not recent) + 2-4 weeks post-change (2025 AHA/ACC Section 5.2.8)	Review results when available. Critical value → immediate action. Only a persistent eGFR decline >30% warrants nephrology referral (Section 5.2.9).
New diuretic started or dose increased	K+, Na+, Creatinine	Baseline + 2-4 weeks post-change (2025 AHA/ACC Section 5.2.8)	Same as above. Watch for hypokalemia (thiazide/loop) and hyponatremia (thiazide).
Routine monitoring (ACEi/ARB/diuretic)	K+, Creatinine, eGFR	Every 6 months	Flag as monitoring gap if overdue (Section 14).

Feedback and Improvement

Recommendation Outcomes

- % of patients at goal within 8 weeks of an executed recommendation
- % of patients reporting a new side effect within 4 weeks of a therapy change
- Clinician approval, modification, and rejection rates

Decision Coverage

- % of decisions resolved by decision tree traversal
- % of decisions requiring RAG (which nodes are most blocking?)
  % of decisions escalated to clinician without recommendation

Safety Signals

- Rate of side effects / abnormal labs / judge overrides / clinician rejections

Judge Calibration

- Agreement matrix vs clinician
- Citation verification pass rate
- Zero-citation recommendation rate

Patient Engagement

- Outreach messages that resulted in a BP reading within 7 days
- Total agent messages THERAPY_ADJUSTMENT to AT_GOAL

Patients Like Me

- Once we have sufficient number of patients, build models to predict which patients may do better on particular meds/combos/dosages than others

Cost Estimate

